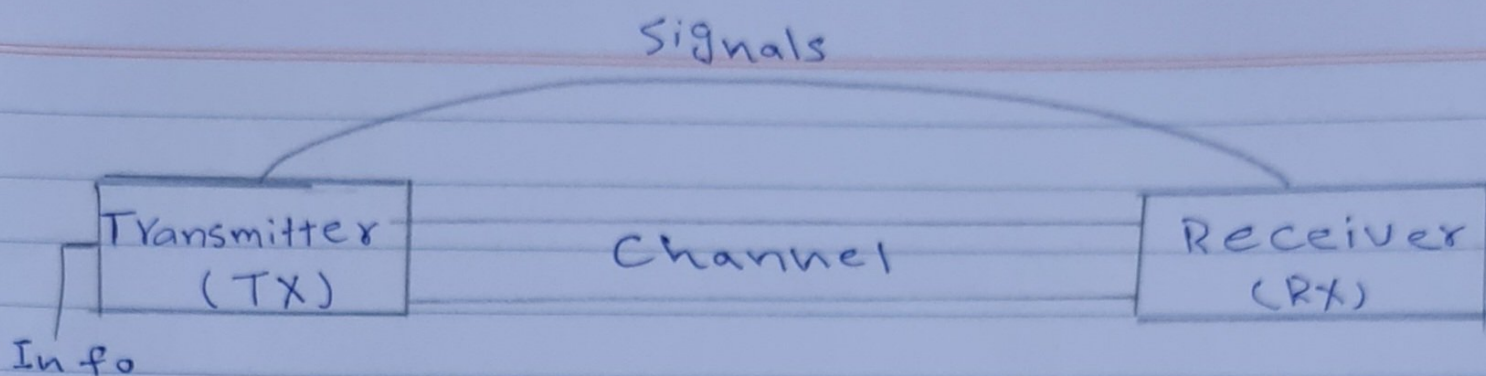




* In order to achieve communication between two systems, a complete communication system must include the following components:

1. Transmitter (TX)
2. Receiver (RX)
3. Signal
4. Channel
5. Information

Transceiver: It is a device that combines both transmitter and receiver

Problem in communication:

1. Noise

{

External noise (outside)

Internal noise (inside)

2. Attenuation

3. Interference.

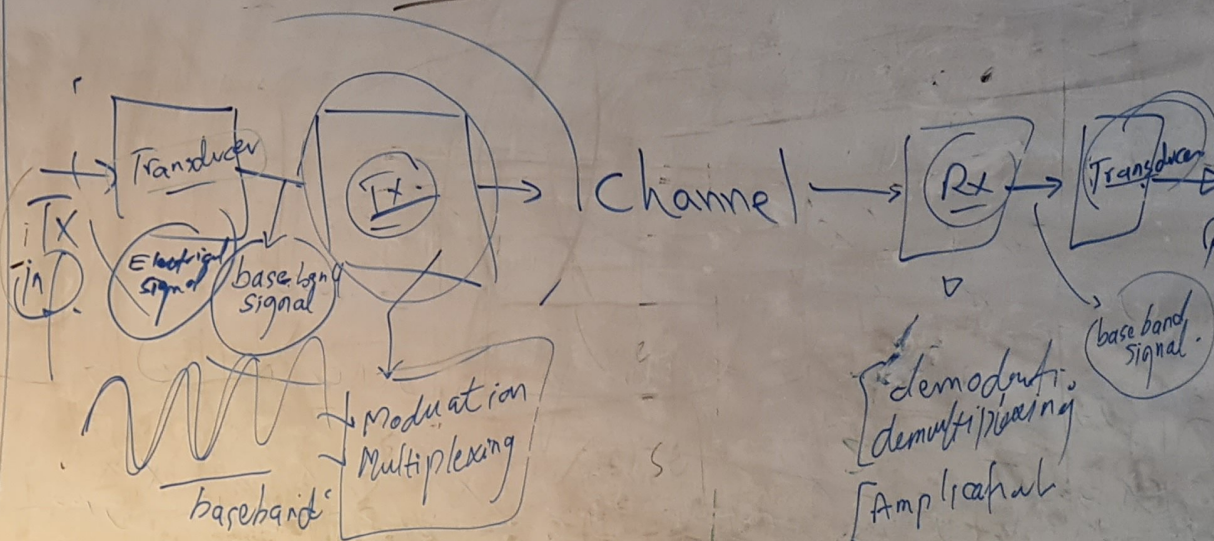
Lecture 1

Main objectives.

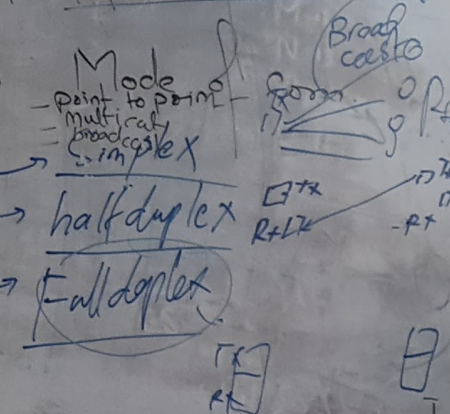
- * System design.
- * Different comm. system type,
- * Digital Analog Sys.

Introduction.

Electrical Communication System



Types of comm.



$$\begin{array}{rcl}
 \frac{\lambda}{4} & = & \frac{1}{4} \times \frac{F}{f} \\
 & & \frac{300 \text{ Hz}}{3400 \text{ Hz}} \\
 & & \frac{3 \times 10^8}{3400} \\
 & & = 22000
 \end{array}$$

Important System factors

1. signals - to - Noise Ratio (SMR) / $SMR = \frac{\text{Power signals}}{\text{Power noise}}$

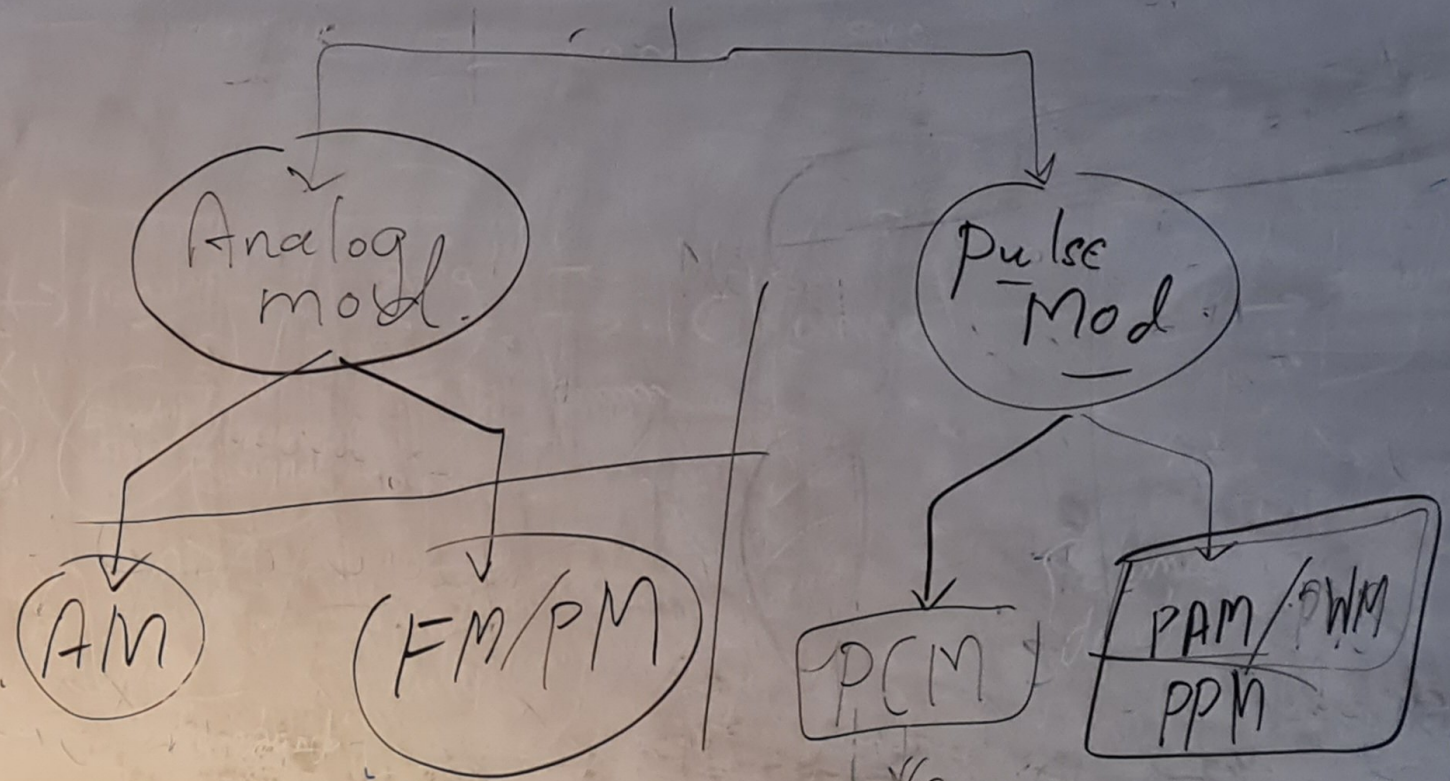
2. Range of Communication

3. Bandwidth

4. Power consumption

Main objectives.

- * System design.
- * Different comm. system type,
- * Digital Analog Sys.



Main objectives.

System design.

Different comm.
System Type,

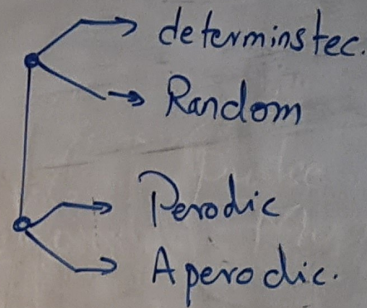
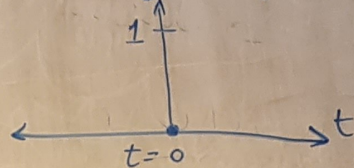
Digital Analog
Sys.

Introduction.

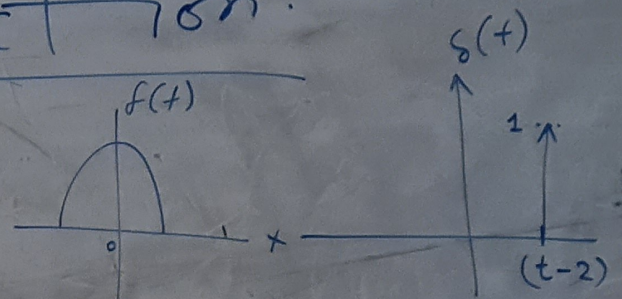
Signals.

A - Types of signal

→ Impulse signal (delta)



$$\int_{-\infty}^{+\infty} \delta(t) \cdot dt = 1$$



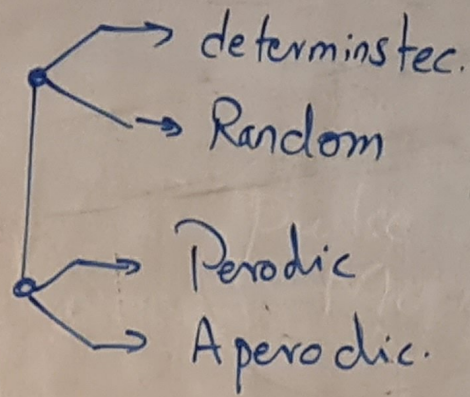
$$\int_{-\infty}^{+\infty} \delta(t) \cdot f(t) \cdot dt = f(0)$$

Lecture II

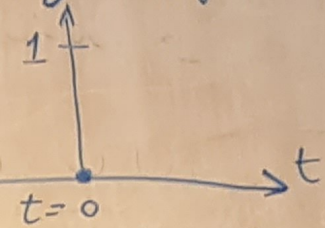
Introduction.

signals.

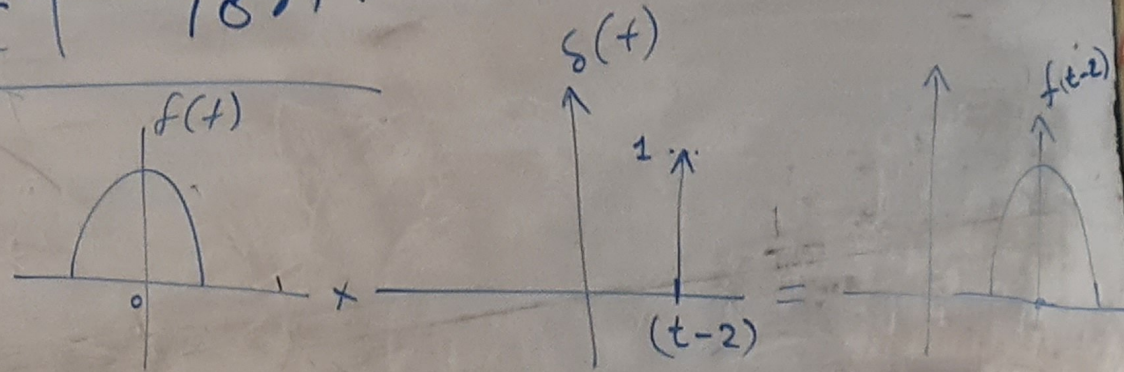
Types of signal



Impulse signal (delta)



$$\int_{-\infty}^{+\infty} \delta(t) \cdot dt = 1 \quad \leftrightarrow \quad \int_{-\infty}^{\infty} \delta(t) \cdot f(t) \cdot dt = f(0)$$



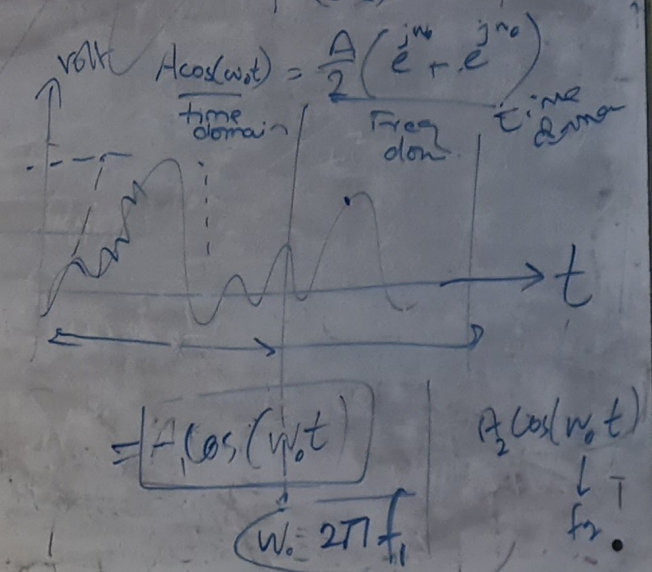
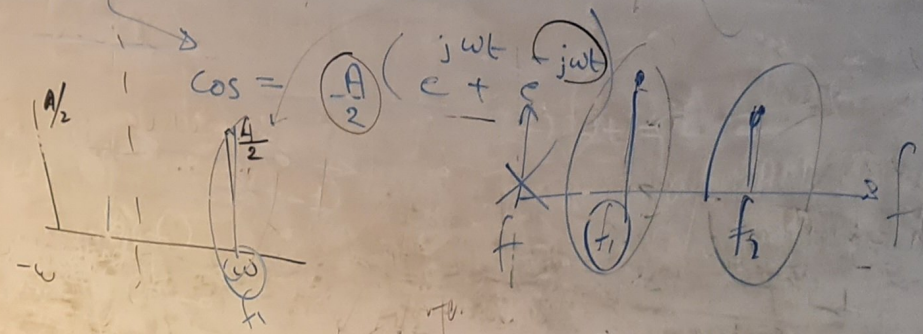
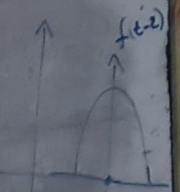
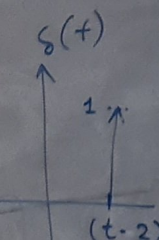
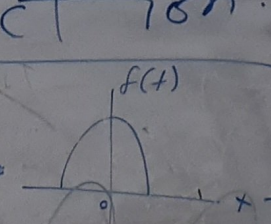
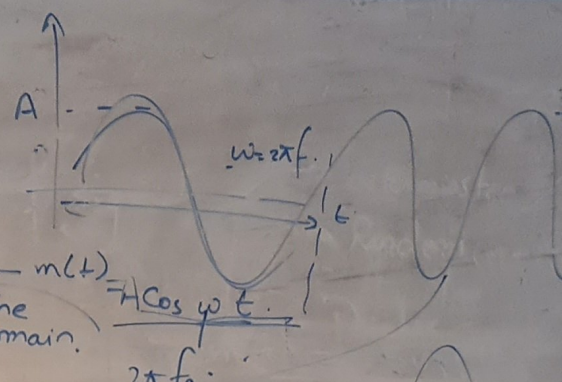
Lecture 1

Introduction

Main objectives.

- * System design.
- * Different comm. system type,
- * Digital modulation.

$$S(t) = e^{j\omega t}$$



$$A \cos(\omega t) = \frac{A}{2} (e^{j\omega t} + e^{-j\omega t})$$

$$= A \cos(\omega t)$$

$$A_2 \cos(\omega_0 t)$$